

AI-Driven Multi-Modal Adaptive Handover
Control for O-RAN
thesis

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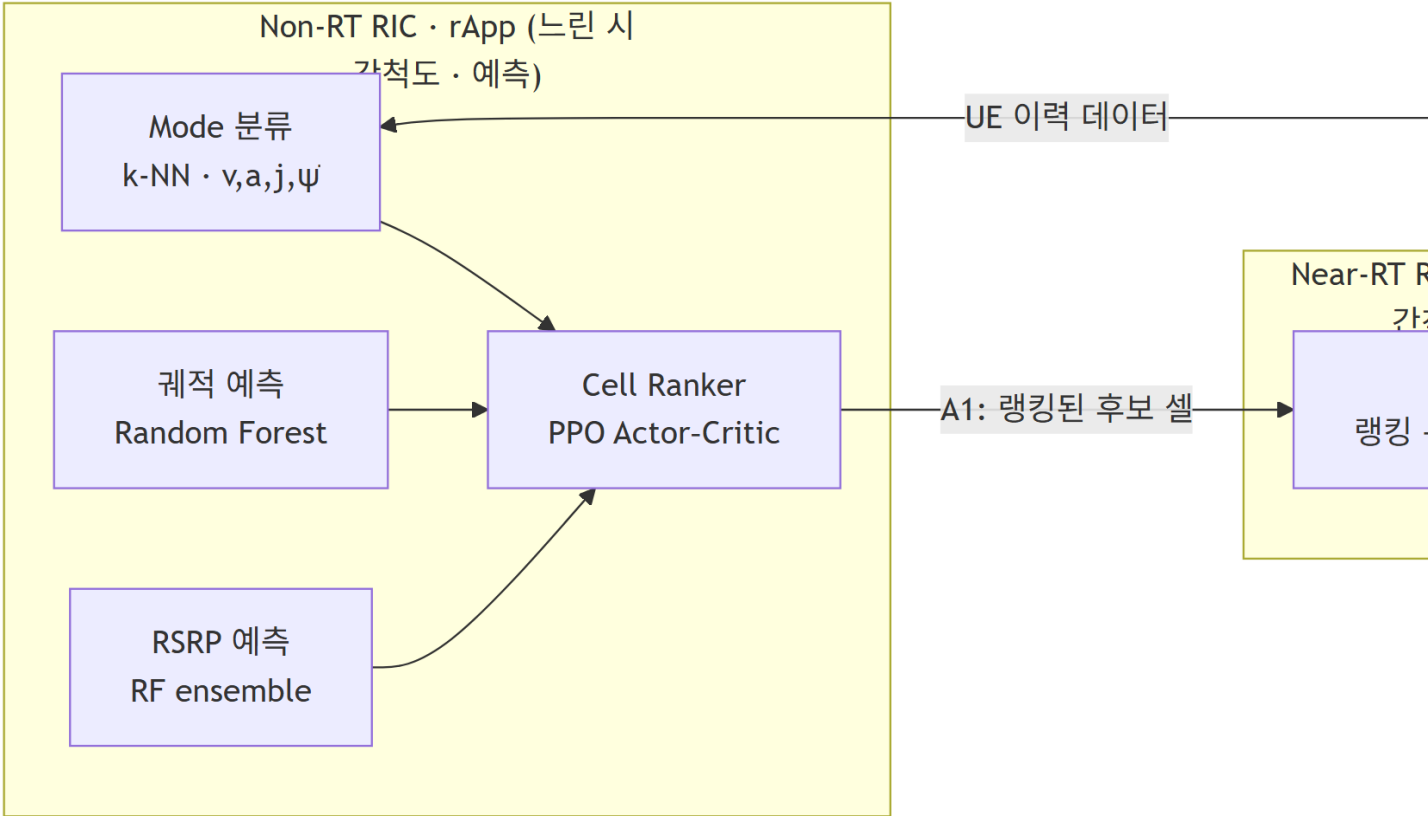
· (7) · RSRP , PPO non-RT RIC **rApp** ,
near-RT RIC **xApp** E2 hierarchical mobility-aware handover .

2 Motivation

-
- (1) **A3 mobility-agnostic** . Traditional A3-event handover TTT / Hysteresis / CIO
· robust , · fading ping-pong handover , late/failed
handover RLF .
- (2) **ML** . · · · **multi-modal** , ML handover
·
- (3) **near-RT** . DQN/LSTM near-RT RIC .
: multi-modal + + O-RAN .
-

3 System Model & Architecture

(rApp, non-RT) (xApp, near-RT) , **A1** .



sub-second non-RT rApp , near-RT xApp
RIC

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4.1 (MC)

$(v, a, j, \dot{\psi}) +$ **k-NN** jerk $j = d^3x/dt^3$ (,) bearing rate $\dot{\psi} = v\kappa$ ()
feature 7 {ped, cyclist, car, bus, train, drone, uav} one-hot $\mathbf{e}(m_t^u)$.

4.2 (TP) · RSRP (P-RSRP)

Random Forest .

- **TP**: (v, a, m, x, y) $\hat{p}_{t+1}^u$
- **P-RSRP**: feature RSRP $\hat{r}_{t+1}^u$

LSTM/DQN “near-RT ” .

4.3 Cell Ranker (CR) — PPO Actor–Critic

- **state:** $s_t^u = [\mathbf{e}(m_t^u), \hat{p}_{t+1}^u, \hat{r}_{t+1}^u]$ — + + RSRP
 - **action:** ()
 - **reward:** $r_t = \alpha w_u R - \beta H - \gamma PP$
 - R : throughput, H : handover penalty, PP : ping-pong penalty
 - clipped surrogate objective(PPO)
- A1 xApp , xApp live E2 load/QoS .

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5.1

MATLAB 5G Toolbox. O-CU 1 / O-DU 3 / O-RU 27, UE 100 , 7 km² . 2.1 GHz, 20 MHz, 46 dBm, UMa path loss, 8 dB shadowing. Fronthaul 10 Gbps (0.1–0.5 ms). Control interval 200 (20) 30 , 95% CI.

: Traditional A3 / O-RAN ML-Assisted(LSTM transfer learning) / Load Balance O-RAN.

5.2 KPI

	MultiModal-AHC	Traditional A3	ML-Assisted	Load Balance
Throughput (Mbps)	77.10 ± 5.72	74.11	78.48	75.23
HO rate	0.0362	0.0962	0.0209	0.4013
Ping-pong (%)	0.00	0.00	12.64	11.29

5.3

(MC)	k-NN	99.84%	
(TP)	Random Forest	RMSE 14.40 m,	1.00
RSRP (P-RSRP)	RF ensemble	RMSE 7.35 dB	
Cell Ranker (CR)	PPO Actor–Critic	reward +23.4, HO failure –18%, 150–180 ep	

stability–agility . A3 TTT ping-pong 0 , · RSRP ping-pong 0 . , throughput baseline CI .

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.

. MC 99.84%, 1.00, RSRP -66.76 vs -66.75 . MATLAB kinematic
 trace , testbed future work .

PPO . Abstract · Sec IV PPO rApp(non-RT) xApp , Sec III-C
 Actor-Critic “real-time decision engine” . non-RT real-time .

. “HO rate = HO failure rate” , MultiModal-AHC HO rate ML-Assisted proactivity
 . “18% failure ” KPI PPO reward . throughput **ping-pong**
 + **HO rate** .

7 & thesis

thesis .

7.1

- **building block** : P-RSRP() TP() thesis channel quality prediction
 · traffic prediction .
- : (v, a, j, ψ) mobility mode thesis (/ AMR /) .
 ped / car / drone .
- **predict-then-decide** : proactive thesis blueprint **Stage 4(chance-constrained PRB)** .

7.2 (foil)

non-RT rApp + near-RT xApp A1/E2/O1 **RIC** . sub-second
 .

thesis .

Handover / mobility	sub-second	RIC (rApp/xApp, A1/E2)	
Importance-aware MAC scheduling	sub-TTI (μ s)	collocation	E2

importance-aware MAC scheduling sub-TTI E2 xApp , GPU-collocated
 RAN+AI(pyAerial + GH200 + cuMAC) . “predictive mobility RIC ” ,

7.3 ()

Predictive mobility RIC (: MultiModal-AHC, $rApp \rightarrow A1 \rightarrow xApp$).
sub-TTI importance-aware MAC scheduling — E2 near-RT (10 ms)
TTI (μs) , GPU-collocated RAN+AI sub-TTI
.

7.4 AI-RAN

predict-then-allocate sub-TTI $\cdot$ PRB $\cdot$ collocation thesis novelty (Stage 3 $\cdot$ 4)
. GH200 + pyAerial(CU $\cdot$ MAC/L1) cuMAC , ROS2 .

8 take-away

1. sub-second RIC .
2. $\cdot$ + + RL predict-then-decide .
3. thesis sub-TTI $\cdot$ PRB , RIC .